

KOTA BHARU, Malaysia

WMO: 486150

Lat: 6.1638N

Lon: 102.3010E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.8	73.1	68.0	103.0	77.1	69.4	108.2	77.9	19.0	81.2	17.9	81.3	2.5	210	0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.0	91.8	80.2	91.3	80.1	90.0	79.7	82.3	89.0	81.6	88.4	81.0	87.8	8.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	158.8	87.0	79.4	152.8	86.0	79.1	150.9	85.7	46.2	88.4	45.3	89.4	44.8	88.3	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	13.8	12.1	69.7	94.3	1.9	1.3	68.4	95.3	67.3	96.1	66.2	96.8	64.9	97.8		
(5)			68.0	84.0	2.3	1.2	66.4	84.8	65.0	85.5	63.7	86.2	62.0	87.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	79.5	80.3	81.6	83.4	83.8	83.2	82.6	82.3	82.0	81.3	80.1	79.7		
	DBStd	2.13	1.77	1.57	1.76	1.63	1.55	1.28	1.43	1.35	1.12	1.53	1.88	1.82		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11559	916	848	981	1003	1048	997	1011	1002	959	971	903	920		
	CDD65	6084	451	428	516	553	583	547	546	537	509	506	453	455		
	CDH74	62916	4063	4134	5476	6541	6882	6157	5876	5609	5135	4932	4054	4057		
	CDH80	21903	826	1094	1882	2719	2956	2544	2297	2157	1894	1601	1059	874		
(14) Wind	WSAvg	5.1	6.7	6.7	5.9	5.2	4.6	4.4	4.2	4.4	4.4	4.2	4.3	6.1		
(15) Precipitation	PrecAvg	104.8	4.7	2.1	4.6	3.8	4.1	4.8	6.3	6.5	7.0	10.7	26.3	22.4		
	PrecMax	146.3	24.4	10.7	26.1	14.9	11.1	11.6	13.6	13.7	15.5	23.5	63.5	60.7		
	PrecMin	60.6	0.0	0.0	0.0	0.0	0.0	0.0	1.8	1.9	2.3	3.1	6.5	1.7		
	PrecStd	21.6	5.0	2.2	5.7	3.5	2.4	2.4	2.8	2.8	3.0	4.6	13.6	12.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.5	88.4	91.0	93.3	93.5	91.8	91.8	91.7	91.0	90.0	88.3	87.8		
		MCWB	78.7	78.1	79.4	80.3	80.7	80.3	79.8	78.8	78.7	79.2	79.2	79.6		
	2%	DB	86.0	87.5	89.5	91.6	91.8	91.2	90.1	90.0	89.7	88.8	87.5	86.2		
		MCWB	77.9	77.9	79.0	80.0	80.8	80.3	79.7	79.0	79.0	79.2	79.3	78.6		
	5%	DB	84.6	86.2	88.2	90.6	91.1	89.9	89.5	89.3	88.6	87.9	86.3	85.2		
		MCWB	77.0	77.4	78.8	79.8	80.7	80.0	79.6	79.1	79.0	79.2	78.9	78.0		
	10%	DB	84.0	85.4	87.5	89.5	89.8	89.4	88.4	88.1	87.8	86.8	85.5	84.2		
		MCWB	76.7	77.1	78.7	79.7	80.1	79.8	79.4	79.1	79.0	78.9	78.6	77.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	80.0	81.3	82.7	83.2	83.3	82.5	81.6	81.6	81.0	81.0	80.6		
		MCDB	84.9	86.5	87.6	90.2	90.3	89.3	88.5	87.6	87.6	87.3	86.0	85.7		
	2%	WB	79.0	79.0	80.1	81.5	82.0	81.9	81.3	80.6	80.6	80.3	80.0	79.5		
		MCDB	84.0	85.5	87.1	89.1	89.3	88.4	87.7	87.3	87.2	86.6	85.3	84.3		
	5%	WB	78.2	78.3	79.6	80.8	81.5	81.1	80.5	80.0	79.9	79.7	79.3	78.9		
		MCDB	83.1	84.4	86.3	88.2	88.6	87.8	87.2	86.6	86.5	85.9	84.3	83.5		
	10%	WB	77.5	77.7	79.0	80.1	80.8	80.4	79.9	79.4	79.3	79.1	78.9	78.2		
		MCDB	82.3	83.5	85.5	87.2	87.8	87.2	86.3	86.0	85.9	85.1	83.8	82.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.8	11.9	12.7	13.0	13.2	13.0	13.1	12.9	11.3	9.4	8.6		
		MCDBR	10.1	12.0	13.0	13.9	14.1	13.9	13.9	14.1	13.8	12.7	11.4	10.1		
	5% WB	MCWBR	4.4	4.8	5.1	5.3	6.0	6.2	6.1	6.0	6.1	5.6	5.0	4.6		
		MCWBR	9.1	10.7	12.0	12.9	13.3	13.3	13.2	13.2	13.0	12.0	10.5	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.464	0.456	0.461	0.445	0.454	0.471	0.476	0.474	0.471	0.466	0.450	0.473		
	taud		2.301	2.306	2.309	2.366	2.350	2.321	2.299	2.290	2.295	2.319	2.365	2.285		
	Ebn at Noon		272	277	275	275	266	259	258	264	268	271	273	266		
	Edh at Noon		43	43	43	40	40	40	42	43	43	42	39	43		
(44) All-Sky Solar Radiation	RadAvg		1427	1758	1853	1871	1691	1627	1613	1676	1662	1469	1261	1209		
	RadStd		115	128	161	93	89	64	94	51	74	97	96	135		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.61	+0.41	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.29	N/A	N/A	+0.67	+0.50	+0.38	N/A	N/A	N/A	N/A

Nomenclature: See separate page